Conditionals, Couplings and Toric Geometry: Kernels, the Coupling Nerve, Cumulants and a Fukaya Programme

Revised draft 4

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Abstract

We organize the conditional and coupling structure of the finite distribution monad D into four layers and separate what is proved from what is conjectural.

(1) *The base.* Markov kernels are the 1-cells of the Kleisli category, and couplings are 2-cells whose vertical composition is the gluing lemma. The natural whiskerings are not functorial, so the couplings do not form a 2-category. Their image under the cost map does: a strict, graded, Lawvere-enriched 2-category.

(2) *The fibre.* The base-relative posterior tower is a functor on the Kleisli category. Its category of elements is a discrete opfibration over it, and every grade of the shadow bounds the distance between transported contexts at every level of the tower.

(3) *The coupling nerve.* Couplings of all arities form the simplicial object $\mathsf{D}(Y^{\bullet+1})$, with marginalization as faces. Every 2-horn fills. The fillers form a product of coupling polytopes, and the maximum-entropy filler is the one whose third face is the vertical composite. For $n \geq 3$ the faces of an n-horn form a cyclic cover, so filling can fail. The failure is measured by the contextual fraction, and $\mathsf{D}(Y^{\bullet+1})$ is not a Kan complex, although $Y^{\bullet+1}$ is.

(4) *Linearization.* Centred couplings are second mixed cumulants, and whiskering defects are the Boolean cumulants of whiskering viewed as a noncommutative expectation. Into the square-zero cone of the circulation ideal, whiskering is an A_∞-functor with exactly two components. Its cohomology keeps only the marginals, and the coupling data are homotopical.

On top of these layers sits the geometry. Coupling polytopes of nondegenerate marginals are Delzant, so they are moment polytopes of toric manifolds whose Kähler class is the product coupling and whose canonical metric restricts to the Fisher metric. Their walls and chambers, monotone cases, Floer theory and Bridgeland stability conditions are treated as before. A new section takes up the five problems the previous draft left open. The parity horns have contextual fraction $1 - 2^{2-n}$ for every n. Whiskering extends to a multi-object A_∞-functor that receives the 2-skeleton of the nerve. Wall-crossing is variation of GIT. Central charges glue continuously across the pentagon–hexagon wall. For generic marginals the product fibre is Floer-trivial (proved). At all sampled points, the Floer-nontrivial fibres lie over the leximin coupling and two signed-balanced couplings (computed). A further section connects the obstruction tower: every horn obstruction is the secondary obstruction of an encoded level-one claim, equal to twice the contextual fraction when all compatible families are admissible, and the Fisher metric

*Code and checks: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/conditionals. Verification scripts are listed in section 14; the worked examples are in the companion [9].

bounds the obstructions from above. A statistical section reads the constructions with synthetic data: the test of independence for a confusion table is the Fisher form at the product coupling, with degrees of freedom equal to the dimension of the coupling polytope; the chamber can be identified from samples with an explicit bound; and two models scored on one label set form a 2-horn whose fillers bound joint error rates. Conjectures are stated with correct types, and a companion paper [9] computes every construction explicitly.

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1 Introduction

Two structures sit on top of the finite distribution monad D. The base-relative posterior tower [4] fixes an outcome base B and solves a *strict integrability* problem: it carries a mixed distributive

law satisfying all four Beck axioms. The conditional/coupling structure [2] lets the base vary along Markov kernels and compares kernels by couplings: it solves a *transport* problem. This paper assembles them, together with a simplicial and a homotopical layer, into one framework.

Layer	Objects and structure	Status
I. Base	finite sets, kernels, coupling 2-cells; the graded shadow	strict 2-category on grades only (section 2)
II. Fibre	the tower over each base; transport along kernels	discrete opfibration; grades act on fibres (section 3)
III. Nerve	$\mathsf{D}(Y^{\bullet+1})$; horns, fillers, boundary fibres	2-horns fill; higher horns obstructed by contextuality (section 4)
IV. Linearization	circulations, cumulants, cones	whiskering is a two-component A_∞-functor into a cone (section 5)
Geometry	coupling polytopes as moment polytopes; stability	proved in the nondegenerate case (sections 6 to 8)
Programme	families over chambers; Fukaya-type functor	conjectural (section 9)

The layers interlock. A 2-cell is a 1-simplex of the nerve, and vertical composition is the canonical filler of an inner 2-horn. The tangent spaces of the filler polytopes are circulations, and those are where the cumulants of layer IV live. The tower of layer II is transported along the 1-cells of layer I, and the grades of layer I bound the transport. The polytopes of layer III are, in the nondegenerate case, moment polytopes, and that is where the geometry enters. What does *not* hold is equally important. The couplings themselves do not form a 2-category, the nerve is not Kan, and the toric picture stops at boundary fibres of 3-way tables.

2 Layer I: kernels, couplings and the graded shadow

D is the finitely supported distribution monad on **Set** and $\mathrm{Kl}(\mathsf{D})$ its Kleisli category. A 1-cell $k\colon B \rightsquigarrow B'$ is a Markov kernel $k\colon B \to \mathsf{D}(B')$, acting on states by $k^{\#}(p)(b') = \sum_b p(b)k(b)(b')$. The results of this section are proved in [2, §7]; we restate them for reference.

Proposition 2.1 (1-cells [2, Props. 7.2, 7.3])**.** *$k^{\#}$ is affine, $(h \circ k)^{\#} = h^{\#}k^{\#}$, $k^{\#} \circ \mu = \mu \circ \mathsf{D}(k^{\#})$, and $\mathrm{TV}(k^{\#}p, k^{\#}q) \le \mathrm{TV}(p, q)$. Attribution sets are transported along k, and precision can only be lost.*

Definition 2.2 (Coupling 2-cells)**.** For 1-cells $f, g\colon X \rightsquigarrow Y$, a coupling 2-cell $\alpha\colon f \Rightarrow g$ is a kernel $\alpha\colon X \to \mathsf{D}(Y \times Y)$ with marginals $f(x)$ and $g(x)$. Its cost is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$; the identity is the diagonal coupling; the vertical composite glues along the common marginal, $(\beta \cdot \alpha)(x)(y, y'') = \sum_{y'} \alpha(x)(y, y')\beta(x)(y', y'')/g(x)(y')$.

Theorem 2.3 (Witnesses and shadow [2, Thms. 7.6, 7.8])**.**

(i) For fixed X, Y, 1-cells and coupling 2-cells form a category under gluing, strictly associative and unital. Cost is subadditive and $\min_\alpha c(\alpha) = d(f, g) = \max_x \mathrm{TV}(f(x), g(x))$.

(ii) Whiskering by a deterministic precomposition is strictly functorial. Whiskering by a stochastic precomposition (mixing couplings) or by a postcomposition (copying on equal pairs, independent on unequal pairs) preserves identities, marginals and the cost bound, but is not functorial.

(iii) Graded 2-cells $f \Rightarrow_\varepsilon g$, $\varepsilon \ge d(f, g)$, with grades adding, form a strict 2-category on kernel 1-cells, and d makes $\mathrm{Kl}(\mathsf{D})$ a Lawvere-enriched category.

Remark 2.4 (Witness layer and shadow). The cost map $\alpha \mapsto c(\alpha)$ sends the witness layer onto the graded shadow. It preserves identities and sends composites to grades no larger than the sum. The shadow is an image of the witness layer, not a substructure of it, and it is the only strict 2-category in sight: with couplings as 2-cells, (ii) rules out a 2-category or a bicategory with these whiskerings. The polytopes $\mathrm{Cpl}(f(x), g(x))$ are the spaces of 2-cells, not further cells; their geometry is the subject of sections 4 and 6.

3 Layer II: the tower as a fibration

The base-relative tower over B has context algebra $A_B = \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathrm{Sh}(B)$, reader comonad $\mathsf{C}^B(Y) = Y \times A_B$ and strict law λ^B [4, §9]. With $\mathrm{Hyp} = \Theta \times \mathsf{D}(-)$, a kernel $k\colon B \to \mathsf{D}(B')$ acts on hypotheses by $\mathrm{Hyp}(k)(\theta, \rho) = (\theta, k^{\#}\rho)$, and we put $A_k = k^{\#} \times \mathsf{D}(k^{\#}) \times \mathsf{D}^2(k^{\#}) \times \mathsf{D}(\mathrm{Hyp}k)$.

Theorem 3.1 (The tower is a functor on the Kleisli category [4, Prop. 9.14, Cor. 9.15, Prop. 9.16])**.**

(i) *A_k is a morphism of D-algebras, preserves coherent contexts, commutes with the canonical section σ, and $A_{h\circ k} = A_h A_k$; for deterministic k it is the earlier base change.*

(ii) *$\mathrm{id} \times A_k$ is a comonad morphism $\mathsf{C}^B \Rightarrow \mathsf{C}^{B'}$ commuting with the laws λ^B, $\lambda^{B'}$.*

(iii) *Admissibility data, coherent lifts and the effective sets are transported; certificates pull back along $f' \mapsto k \rhd f'$; failure levels can only rise and the secondary obstruction can only fall.*

(iv) *Conditioning on an outcome observed after the channel equals conditioning on the pulled-back predicate $k \rhd 1_{e'}$, then changing base.*

Definition 3.2 (The total category)**.** Let $\mathcal{E}$ be the category whose objects are pairs (B, z) with B a finite set and $z \in A_B^{\mathrm{coh}}$ a coherent context, and whose morphisms $(B, z) \to (B', z')$ are kernels k with $A_k(z) = z'$. Let $\pi\colon \mathcal{E} \to \mathrm{Kl}(\mathsf{D})$ forget z.

Proposition 3.3 (Split opfibration)**.** *π is a discrete opfibration, and in particular a split opfibration. The cocartesian lift of $k\colon B \to B'$ at (B, z) is $k\colon (B, z) \to (B', A_k z)$. Through $\mathrm{id} \times A_k$, the comonads C^B and the laws λ^B are carried along the lifts.*

Proof. $\mathcal{E}$ is the category of elements of the functor $B \mapsto A_B^{\mathrm{coh}}$, $k \mapsto A_k$, which is a functor by theorem 3.1(i); categories of elements of set-valued functors are discrete opfibrations. The statement about comonads and laws is theorem 3.1(ii). □

Proposition 3.4 (Grades act on fibres)**.** *Let $f, g\colon B \to \mathsf{D}(B')$ and let $z = (\rho, \Pi, \Omega, \Sigma)$ be a coherent context. Put* TV *on $\mathsf{D}B'$, the Kantorovich metric W_1 over* TV *on D^2B', the Kantorovich metric over that on D^3B', and on hypotheses the metric that is* TV *of predictives for equal descriptors and* 1 *otherwise. Then every component of $A_f(z)$ and $A_g(z)$ is within $d(f, g)$. So a graded 2-cell $f \Rightarrow_\varepsilon g$ gives ε-close transported contexts at every level.*

Proof. $\mathrm{TV}(f^{\#}\rho, g^{\#}\rho) \le \sum_b \rho(b)\mathrm{TV}(f(b), g(b)) \le d(f, g)$ by convexity. For Π, push Π forward along $\rho \mapsto (f^{\#}\rho, g^{\#}\rho)$. This is a coupling of $\mathsf{D}(f^{\#})\Pi$ and $\mathsf{D}(g^{\#})\Pi$ of cost $\mathbb{E}_\Pi \mathrm{TV}(f^{\#}\rho, g^{\#}\rho) \le d(f, g)$. Level three and the shadow are the same argument one level up. The companion verifies the bound in a worked example, where all three levels give $\frac{2}{15} \le \frac{1}{6}$ [9, §11]. □

The proof of theorem 3.1 uses one fact beyond naturality: Kleisli extension is a morphism of free D-algebras. All other components of A_k are pushforwards along functions. So the strict integrability of each fibre and the transport between fibres are compatible.

Ordered fibres: support policies. A decision layer needs sets of laws ordered by strength, not single contexts [7, Section 14]. At level one this is a posetal enlargement of $\mathcal{E}$.

Definition 3.5 (The ordered total category)**.** Let $\mathcal{K}(B)$ be the closed convex subsets of $\mathsf{D}B$ ordered by inclusion, with $k\colon B \to \mathsf{D}(B')$ acting by images $k^{\#}$. Let $\mathcal{E}_{\leq}$ have objects (B, L) with $L \in \mathcal{K}(B)$ and an arrow $(B, L) \to (B', L')$ for each kernel k with $k^{\#}L \subseteq L'$.

Proposition 3.6 (Ordered fibres)**.** *The projection $\mathcal{E}_{\leq} \to \mathrm{Kl}(\mathsf{D})$ is an opfibration whose fibres are the posets $\mathcal{K}(B)$, with cocartesian lifts $k\colon (B, L) \to (B', k^{\#}L)$. Singletons $\{\rho\}$ with ρ the level-one component of a coherent context embed the level-one part of $\mathcal{E}$ (proposition 3.3). Policies $L \mapsto L \cap \{\rho \geq \tau\}$ and floor summaries are an idempotent comonad and an idempotent monad on each fibre; they do not commute with transport, and the comparisons are inclusions: floors push forward soundly, policies imposed before transport certify more, and floor summaries form a normal oplax functor [4, Prop. 10.23, Prop. 10.24]. Along a chain of transports and vertical steps, certification is path-independent and is lost only at summaries [4, Thm. 10.26].*

Proof. $(h \circ k)^{\#} = h^{\#}k^{\#}$, so $B \mapsto \mathcal{K}(B)$ is a functor into posets, and its category of elements is an opfibration with these fibres. Singletons are preserved by $k^{\#}$. The rest is proved in the cited results. □

A fibred comonad on the discrete opfibration $\mathcal{E}$ itself would be trivial, since a counit inside a discrete fibre is an identity. This is why the tower's comonads are carried *alongside* $\mathcal{E}$, as the family C^B joined by the comonad morphisms of theorem 3.1(ii), and why a policy can be a non-trivial comonad only on ordered fibres.

4 Layer III: the coupling nerve

Definition 4.1 (The coupling nerve)**.** For a finite set Y, let $N_k(Y) = \mathsf{D}(Y^{k+1})$, with face maps d_i that marginalize out the ith coordinate and degeneracies s_i that push forward along the diagonal repeating it. Since $Y^{\bullet+1}$ is a simplicial set (the Čech nerve of $Y \to 1$) and D is a functor, $N(Y) = \mathsf{D}(Y^{\bullet+1})$ is a simplicial set. The relative version over a set X has k-simplices the kernels $X \to \mathsf{D}(Y_0 \times \cdots \times Y_k)$, with all structure taken pointwise in x.

A 1-simplex of $N(Y)$ is a coupling of its two vertices. A coupling 2-cell $\alpha\colon f \Rightarrow g$ between kernels $X \rightsquigarrow Y$ is an X-indexed 1-simplex with vertices f and g. A horn Λ^n_k in N is a compatible family of laws on the n faces other than the kth, and a filler is an n-simplex with those faces. In probabilistic terms a horn is an empirical model on the cover of $\{0, \dots, n\}$ by those faces, and a filler is a global section [24].

Theorem 4.2 (2-horns fill, and composition is the canonical filler)**.** *Let p_{01} and p_{12} be compatible laws on $Y_0 \times Y_1$ and $Y_1 \times Y_2$ with common marginal p_1, the data of an inner horn Λ^2_1.*

(i) The fillers form the product, over x_1 with $p_1(x_1) > 0$, of the transportation polytopes of the slices $p(\cdot, x_1, \cdot)$ with marginals $p_{01}(\cdot, x_1)$ and $p_{12}(x_1, \cdot)$. If every slice has full support and nondegenerate marginals, the filler polytope is Delzant.

(ii) The gluing filler $p^{\mathrm{gl}} = p_{01}\, p_{12}/p_1$ is the unique filler of maximal entropy, and its face d_1 is the vertical composite of the two couplings.

(iii) Any filler differs from p^{gl}, slice by slice, by $p_1(x_1)\,\mathrm{Cov}(1_{\{x_0\}}, 1_{\{x_2\}} \mid x_1)$, a circulation of the slice.

The outer horns Λ^2_0 and Λ^2_2 fill in the same way, by gluing over the shared vertex.

Proof. (i) Each marginal constraint involves one value of x_1, so the constraints decouple into slices. Each slice is a two-way transportation polytope, Delzant under the stated conditions (theorem 6.4), and a product of Delzant polytopes is Delzant with respect to the product lattice. (ii) $\log p^{\mathrm{gl}}$ is a sum of a function of (x_0, x_1) and a function of (x_1, x_2), so it is orthogonal to every tangent vector of the filler polytope. So p^{gl} is a critical point of the strictly concave entropy on the polytope, hence its unique maximizer. Equivalently, $H(X_0X_1X_2) \le H(X_0X_1) + H(X_1X_2) - H(X_1)$, with equality exactly under conditional independence. Summing p^{gl} over x_1 gives the gluing formula of definition 2.2. (iii) is a direct computation. □

So the 2-cells of layer I are the 1-simplices of the nerve, and their vertical composition is the canonical filling of inner 2-horns. The other fillers are conditional cumulants, which is the link to layer IV.

Theorem 4.3 (Higher horns)**.** *Let $n \ge 3$.*

(i) The n faces of Λ^n_k, viewed as subsets of $\{0, \dots, n\}$, form a cover that is not acyclic. For $n = 2$ it is acyclic.

(ii) Hence, for $|Y| \ge 2$, there are compatible horn data in $N(Y)$ with no filler.

(iii) For compatible horn data, the minimal unmodeled weight with the deterministic global assignments as sources, each face weighted $1/n$, equals the contextual fraction of the horn, and it vanishes exactly when the horn fills. For the parity horns on two outcomes (uniform on assignments of fixed parity on each face, with exactly one face odd) it is $1 - 2^{2-n}$ for every n-horn.

Proof. (i) Use Graham's reduction for acyclic hypergraphs [25]: repeatedly delete vertices in a single edge and edges contained in others; the hypergraph is acyclic if and only if this empties it. In Λ^n_k the vertex k lies in all n faces and every other vertex lies in $n-1$ of them. No face contains another, so for $n \ge 3$ no reduction step applies. For $n = 2$ the two faces share one vertex and reduce away. (ii) By Vorob'ev's theorem [36], every compatible family extends to a global law exactly on acyclic covers. Data on two outcomes embed into any Y with $|Y| \ge 2$. (iii) A filler is a global section, so this is [2, Thm. 7.12] applied to the horn's cover. For the parity horn on bits $x_0, \dots, x_n$, put $\sigma = \sum_j x_j$ and let $p_i \in \{0, 1\}$ be the parity of face $i \ne k$. A global assignment meets every face parity if and only if $x_i = \sigma - p_i$ for all $i \ne k$; then x_k is fixed by $\sum_j x_j = \sigma$, and this is consistent because $\sum_i p_i = 1$. So exactly two global assignments, one for each σ, are compatible with all faces, and they differ on every face. Every other deterministic source puts mass on an assignment of probability zero on some face. On each face the two compatible sources have distinct restrictions of probability 2^{1-n}, so each can carry weight at most 2^{1-n}; the noncontextual fraction is 2^{2-n}. Exactly verified certificates for $n \le 6$ and the count for $n \le 8$ are in `check_horns_n.py`. □

Corollary 4.4 (The nerve is not Kan)**.** *For $|Y| \ge 2$, $N(Y) = \mathsf{D}(Y^{\bullet+1})$ fills every 2-horn but not every n-horn for $n \ge 3$. In particular it is not a Kan complex, although $Y^{\bullet+1}$ is. The monad D does not preserve horn filling, because it does not turn compatible families over cyclic covers into global laws, and the obstruction is the contextual fraction.*

Proposition 4.5 (Boundary fibres)**.** *Prescribing all three faces of a 2-simplex, that is all three 2-margins of a 3-way table on $Y_0 \times Y_1 \times Y_2$, gives a planar 3-way transportation polytope of dimension $(n_0-1)(n_1-1)(n_2-1)$ when it is nonempty and generic. It can be empty even for compatible faces, because the cover by the three edges of a triangle is cyclic.*

(i) For $n_0 = 2$ it is a capacitated two-way transportation polytope, whose constraint matrix is totally unimodular, so it is Delzant whenever it is simple.

(ii) Every rational polytope is a planar $r \times s \times 3$ transportation polytope [26], so boundary fibres are not toric in general. Already for $3 \times 3 \times 3$, generic instances are simple but not smooth: they are moment polytopes of toric orbifolds [29], not of manifolds [9, §10].

Proof. For $n_0 = 2$ the second layer is $m - y$, where y is the first layer, m is the $(1,2)$-margin, and y has fixed row and column sums with $0 \le y \le m$. A bipartite incidence matrix stacked with an identity is totally unimodular, so edges are ± 1 cycle vectors and the argument of theorem 6.4 applies at simple vertices. (ii) combines the cited theorem with the computation in the companion. □

So the toric regime of the nerve is exactly the 1-simplices and the 2-horns. Boundary fillings leave it, and higher horns can be empty.

Gluing under a support floor. A support policy asks more of a filler than existence. With a floor τ on every cell of the global table, three statuses separate: a filler exists; a filler meets the floor; the maximum-entropy filler is among those that do. A new obstruction appears between the first two.

Definition 4.6 (Support-contextuality)**.** A compatible family on a cover is *support-contextual at floor τ* if it glues, every local table meets the local floor implied by τ (each local cell at least τ times the number of global cells over it), and no global section has every cell at least τ.

Proposition 4.7 (The chain)**.** *For binary $X-Y-Z$ with fixed (X,Y) and (Y,Z) tables π_{XY}, π_{YZ}, a filler with all eight cells at least τ exists iff $\tau \le \tau_{\text{any}} = \frac{1}{2}\min(\min \pi_{XY}, \min \pi_{YZ})$, and the conditional product, the maximum-entropy filler with $H(X,Y,Z) = H(X,Y) + H(Y,Z) - H(Y)$, meets the floor iff $\tau \le \tau_{CI} = \min_{x,y,z} \pi_{XY}(x,y)\pi_{YZ}(y,z)/\pi_Y(y)$. Always $\tau_{CI} \le \tau_{\text{any}}$. A chain is never support-contextual.*

Proof. Given $Y = y$ the filler is a 2×2 coupling of $\pi_{XY}(\cdot, y)$ and $\pi_{YZ}(y, \cdot)$; with all cells floored it is the coupling polytope of the reduced margins (proposition 6.9), non-empty iff each reduced margin is non-negative, that is $\pi_{XY}(x,y) \ge 2\tau$ and $\pi_{YZ}(y,z) \ge 2\tau$. The conditional product is one filler, so $\tau_{CI} \le \tau_{\text{any}}$; its cells are the displayed products. The local floor is the same condition as $\tau \le \tau_{\text{any}}$. □

Theorem 4.8 (The binary triangle under a floor)**.** *Let binary X_0, X_1, X_2 carry consistent tables on the contexts $\{0,1\}, \{1,2\}, \{2,0\}$, with correlations $E_{ij} = \mathbb{E}[(-1)^{X_i + X_j}]$. For $c \in \{0,1\}^3$ let $\bar{c}$ be its complement and*

$$A_c = 1 - \tfrac{1}{2} \sum_{\{i,j\}} P\big(X_i \oplus X_j \ne c_i \oplus c_j\big) = \tfrac{1}{4}\Big(1 - \sum_{\{i,j\}} \gamma_{ij} E_{ij}\Big), \qquad \gamma_{ij} = -(-1)^{c_i + c_j},$$

a function of the pairwise tables. Then a global section with every cell at least $\tau \ge 0$ exists iff every pairwise cell is at least 2τ and $A_c \ge 2\tau$ for the four pairs $\{c, \bar{c}\}$; equivalently

$$\sum_{\{i,j\}} \gamma_{ij} E_{ij} \le 1 - 8\tau \qquad \text{for every } \gamma \text{ with an odd number of } -1\text{s}.$$

At $\tau = 0$ these are the 3-cycle noncontextuality inequalities [8]. The family is support-contextual at τ exactly when every pairwise cell is at least 2τ, all four inequalities hold at $\tau = 0$, and one fails at τ.

Proof. The pairwise-margin map on $\mathbb{R}^{\{0,1\}^3}$ has rank 7 and kernel spanned by $\chi(c) = (-1)^{c_0+c_1+c_2}$, so the global sections of a gluable family are $J + s\chi$ for s in an interval. A floored section exists iff $\max_s \min_c(J_c + s\chi_c) \geq \tau$. With four slopes $+1$ and four -1 the maximum of the minimum is $\frac{1}{2}\min\{J_c + J_{c'} : \chi(c) = 1, \chi(c') = -1\}$, and these sums do not depend on s. Cells of opposite parity differ in one coordinate or in all three. If in one, $J_c + J_{c'}$ is a cell of the context of the other two variables. If in all three, $c' = \bar{c}$ and $J_c + J_{\bar{c}}$ is the probability that the variables, after flipping those with $c_i = 1$, are all equal. A non-constant binary triple disagrees on exactly two of the three pairs, so this probability is $1 - \frac{1}{2}\sum P(\text{disagree}) = A_c$; with $P(X_i \oplus X_j \neq e) = \frac{1}{2}(1 - (-1)^e E_{ij})$ it is the displayed affine form, and $A_c \geq 2\tau$ is the stated inequality. The sign vectors γ have product -1, an odd number of -1s. The case $\tau = 0$ is gluability, and the last sentence is definition 4.6 with local floor 2τ. □

Example 4.9 (A support-contextual triangle)**.** Let the global table put 0 on 000 and 111 and $\frac{1}{6}$ on the other six cells, and give the cover its three pairwise tables. Every pairwise cell is $\frac{1}{6}$ or $\frac{1}{3}$, so the local floor holds for every $\tau \leq \frac{1}{12}$, and the family glues. But $A_{000} = J_{000} + J_{111} = 0$ on every global section, so no global section meets any positive floor: the family is support-contextual at every $\tau \in (0, \frac{1}{12}]$. The other three A_c equal $\frac{1}{3}$. In the pre-registered experiment of [7, Section 14], 124 of 519 random gluable triangles were support-contextual at some tested floor.

5 Layer IV: cumulants and the square-zero cone

Proposition 5.1 (Couplings are second cumulants)**.** *For $\alpha \in \mathrm{Cpl}(f, g)$, $\alpha - f \otimes g$ is the cross-covariance $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$ of the indicator vectors, a circulation. The product coupling is the point where all mixed cumulants vanish, and it is the only point of rank one, the Birch point of the independence model [30]. For level-two couplings of second-order states,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \mathrm{Cov}_\Pi(\rho, \rho'),$$

so flattening keeps only the second mixed cumulant of the two random distributions.

Fix $f \in \mathsf{D}(Y)$ and let A_f be the matrices on $Y \times Y$ whose row and column sums are multiples of f. With the gluing product $a \cdot b = a\,\mathrm{diag}(f)^{-1} b$ and unit $\mathrm{diag}(f)$, A_f is an associative unital algebra, spanned by the couplings of f with itself. The circulations I_f form a two-sided ideal, and $A_f / I_f \cong \mathbb{R}$ records the total mass. Whiskering by a kernel h is a linear unital map $h_* \colon A_f \to A_{h\#f}$, which need not be multiplicative: a noncommutative expectation in the sense of homotopy probability theory [27, 28]. Its Boolean cumulants [31] are $b_1 = h_*$, $b_2(a, b) = h_*(a \cdot b) - h_* a \cdot h_* b$, and so on over interval partitions. The whiskering defect of theorem 2.3(ii) is $-b_2$.

Theorem 5.2 (Whiskering is a two-component A_∞-functor)**.** *Let $C = A_{h\#f} \oplus \varepsilon I_{h\#f}$, with ε of degree -1, $\varepsilon^2 = 0$ and $d(\varepsilon x) = x$. Then C is a differential graded algebra with $H^0(C) = A_{h\#f}/I_{h\#f} \cong \mathbb{R}$ and $H^{-1}(C) = 0$: it is the mapping cone of $I \hookrightarrow A$. The maps*

$$F_1 = h_*, \qquad F_2 = \varepsilon\, b_2, \qquad F_n = 0 \ (n \geq 3)$$

form an A_∞-functor $A_f \to C$.

Proof. Every b_2 is a circulation, because both composites couple the same marginals, so F_2 lands in εI. C is the trivial extension of A by the bimodule εI, and d is a derivation because I is an ideal. Arity two of the functor equation is $dF_2 = b_2$, true by definition. In arity three, with $F_3 = 0$, it

reads $b_2(ab,c) - b_2(a,bc) = h_*a \cdot b_2(b,c) - b_2(a,b) \cdot h_*c$. Using only linearity of h_* and associativity of gluing,

$$\begin{aligned} b_2(ab,c) - b_2(a,bc) &= \big[h_*(abc) - h_*(ab)\,h_*c\big] - \big[h_*(abc) - h_*a\,h_*(bc)\big] \\ &= h_*a\,h_*(bc) - h_*(ab)\,h_*c, \\ h_*a \cdot b_2(b,c) - b_2(a,b) \cdot h_*c &= \big[h_*a\,h_*(bc) - h_*a\,h_*b\,h_*c\big] - \big[h_*(ab)\,h_*c - h_*a\,h_*b\,h_*c\big] \\ &= h_*a\,h_*(bc) - h_*(ab)\,h_*c. \end{aligned}$$

So the identity holds for all a, b, c and every finite Y; it says that b_2 is the Hochschild coboundary of h_*. In arity n the equation has terms $F_{n-1}(\dots, ab, \dots)$, $m_1 F_n$ and $F_i \cdot F_j$ with $i+j=n$. For $n \geq 5$ each involves some F_k with $k \geq 3$, so all vanish. For $n = 4$ the only survivor is $F_2 \cdot F_2 \in \varepsilon^2 I = 0$. The script checks the identity on random inputs only as a unit test of the code. □

Remark 5.3. The gluing algebra splits as $A_f \cong \mathbb{R} \times M_{n-1}(\mathbb{R})$, with the product coupling and $\mathrm{diag}(f) - f \otimes f$ as central idempotents. The functor of theorem 5.2 is the only one with first component h_*, for every mass-preserving linear map in place of h_*, and up to homotopy it is the mass [1]. So the coupling information is chain-level.

Remark 5.4. The proof does not use the general theory of [27, 28], where the higher components of the cumulant morphism are the higher cumulants, which are nonzero here. Dropping components of an A_∞-morphism does not in general give an A_∞-morphism. What makes $F_{\geq 3} = 0$ work is the square-zero target.

Remark 5.5. This settles, in the minimal model, the question of the previous draft whether whiskering is an A_∞-functor. The higher Boolean cumulants are nonzero [9, §9], but they are not needed here. They appear as higher components only in larger models such as the free cone, as in the bar-construction description of cumulants. The defects span all of I for three and four points [9, §10], so H^0 of the cone generated by the defects alone is also A/I.

The four layers together. Under the cone, all coupling information is homotopical: H^0 keeps only masses, so the 1-cells of layer I, while the couplings live in degree -1, in the circulations. These are the tangent spaces of the polytopes of layer III. The cumulants of proposition 5.1, the defects of theorem 5.2 and the deviations of 2-horn fillers from the canonical filler (theorem 4.2(iii)) are all circulations, and the Fisher metric of proposition 6.5 measures them. Layer II carries the tower along the same kernels, with the grades of layer I as bounds.

6 Coupling polytopes are moment polytopes

6.1 The polytope and its lattice

Let $f \in \mathsf{D}(Y)$ and $g \in \mathsf{D}(Y')$ have full support, with $|Y| = n$, $|Y'| = m$. (For coupling 2-cells $Y = Y'$; rectangular polytopes arise when gluing contexts [2, Prop. 7.11].) The *coupling polytope* is

$$\mathrm{Cpl}(f,g) = \Big\{ \alpha \in \mathbb{R}_{\geq 0}^{Y \times Y'} : \ \textstyle\sum_{y'} \alpha(y,y') = f(y), \ \sum_{y} \alpha(y,y') = g(y') \Big\}.$$

Its tangent space is the space of *circulations*, matrices with zero row and column sums. The integer circulations form a lattice Λ of rank $(n-1)(m-1)$; a basis is $E_{ij} - E_{i,m} - E_{n,j} + E_{n,m}$ for $i < n$, $j < m$.

Proposition 6.1 (Interior and Fisher metric)**.** $\mathrm{Cpl}(f,g)$ *has dimension* $(n-1)(m-1)$*, its relative interior is the set of strictly positive couplings, and on it the Fisher metric* $\mathcal{I}_\alpha(v,w) = \sum v(y,y')w(y,y')/\alpha(y,y')$ *is positive definite on circulations.*

Proof. The product coupling $f \otimes g$ is strictly positive, so no inequality $\alpha \geq 0$ is an implicit equality, and the relative interior of the polytope in its affine hull is $\{\alpha > 0\}$. There $\mathcal{I}_\alpha(v,v) = \sum v^2/\alpha > 0$ for $v \neq 0$. □

Remark 6.2. The first draft argued nondegeneracy through a radical and then distinguished "the Fisher metric" from "the Fisher–Rao metric". These are the same metric, the Fisher information metric. The square-root embedding is a way of computing its geodesic distance [2, Prop. 8.1].

6.2 Nondegeneracy implies Delzant

Definition 6.3 (Walls)**.** The pair (f,g) is *degenerate* if $\sum_{y\in I} f(y) = \sum_{y'\in J} g(y')$ for some nonempty proper subsets $I \subsetneq Y$ and $J \subsetneq Y'$. These equations define hyperplanes, the *walls*, in the space of pairs of marginals; the *chambers* are the connected components of the complement of the walls.

Theorem 6.4 (Coupling polytopes are Delzant)**.** *If* (f,g) *is nondegenerate, then* $\mathrm{Cpl}(f,g)$ *is a Delzant polytope with respect to* Λ*: it is simple, its facet normals are rational, and at each vertex the primitive edge vectors form a basis of* Λ*. Its edges at a vertex are the fundamental cycles of a spanning tree.*

Proof. Regard α as a weighting of the edges of the complete bipartite graph on $Y \sqcup Y'$. *Vertices are forests.* If the support of a feasible α contains a cycle, adding $\pm\epsilon$ alternately around it stays feasible, so α is not a vertex. If the support is a forest, the marginal equations determine α on it uniquely (peel off leaves), so α is a vertex. *Nondegeneracy makes them spanning trees.* Every node has positive marginal and hence an incident edge of the support. If the support forest had two or more components, take one with row set I and column set J. Then $\sum_I f = \sum_J g$; I and J are nonempty, and if $I = Y$ then $\sum_J g = 1$ forces $J = Y'$ and the component is everything. So I and J are proper, contradicting nondegeneracy. *Simplicity and smoothness.* At a vertex with spanning tree T, the zero cells are the $nm - (n+m-1) = (n-1)(m-1)$ non-tree cells c. Each closes a unique cycle in $T \cup \{c\}$, giving a circulation C_c with entries in $\{0, \pm 1\}$ and $C_c(c) = 1$. A circulation w equals $\sum_{c\notin T} w(c)C_c$, because the difference is a circulation supported on the tree, hence zero. So the feasible directions at the vertex form the cone $\{\sum_c t_c C_c : t \geq 0\}$ on exactly $\dim \mathrm{Cpl}(f,g)$ independent generators (simplicity), and the C_c are integral and form a $\mathbb{Z}$-basis of Λ, since integer coefficients $w(c)$ suffice (smoothness). The facet normals are the coordinate functionals $\alpha \mapsto \alpha(c)$, which are rational. □

By Delzant's theorem [15] there is a compact connected toric symplectic manifold $(X_{f,g}, \omega)$ of real dimension $2(n-1)(m-1)$, unique up to equivariant symplectomorphism, whose moment map $\mu\colon X_{f,g} \to \mathrm{Cpl}(f,g)$ has image the coupling polytope. The fibre over an interior coupling is a Lagrangian torus $T^{(n-1)(m-1)}$, and the fibre over a vertex is a point.

Proposition 6.5 (Fisher metric and Guillemin's potential)**.** *Suppose every inequality* $\alpha(c) \geq 0$ *defines a facet of* $\mathrm{Cpl}(f,g)$*. The canonical Kähler metric of* $X_{f,g}$ *constructed by Guillemin [19] has, in action–angle coordinates over the interior, the form* $\mathrm{Hess}\, G$ *in the polytope directions, with symplectic potential*

$$G(\alpha) = \tfrac{1}{2} \sum_c \alpha(c) \log \alpha(c).$$

Hence the Fisher metric on $\mathrm{Cpl}(f,g)$ *is twice the restriction of the canonical Kähler metric to the real section, and the product coupling* $f \otimes g$*, the coupling of maximal entropy, is the unique critical point of* G *on the polytope.*

Proof. Guillemin's potential is $\frac{1}{2}\sum_i \ell_i \log \ell_i$ over the facets, where $\ell_i \geq 0$ are the facet inequalities written with primitive inward normals. Here $\ell_c(\alpha) = \alpha(c)$, and the coordinate functional is primitive on Λ because some integer circulation has entry 1 at c. Then $\operatorname{Hess} G(v,v) = \frac{1}{2}\sum_c v(c)^2/\alpha(c)$. The critical-point condition is that $\log \alpha$ is orthogonal to all circulations, that is $\log \alpha(y,y') = r(y)+s(y')$, which together with the marginals gives $\alpha = f \otimes g$. Uniqueness follows from strict convexity. □

If some cell is not a facet, for instance when $f(y) + g(y') > 1$ forces $\alpha(y,y') > 0$, the potential omits that term and the identity holds only up to those terms.

6.3 Walls, chambers and monotone cases

Proposition 6.6 (Chambers)**.** *The set of spanning trees that support vertices, and hence the combinatorial type of* $\mathrm{Cpl}(f,g)$ *and the toric manifold* $X_{f,g}$ *up to equivariant diffeomorphism, is constant on each chamber. The type can change across a wall. For* $f = (\frac{1}{2}, \frac{1}{2})$ *and* $g = (t, \frac{2}{3} - t, \frac{1}{3})$ *the polytope is a hexagon for* $t < \frac{1}{2}$ *and a quadrilateral for* $t > \frac{1}{2}$*, both Delzant: two facets become redundant across the wall* $g_1 = f_1$*.*

Proof. The tree solution on a spanning tree T assigns to each tree edge the value $\sum_I f - \sum_J g$ (up to sign) for the bipartition (I,J) obtained by deleting that edge. So feasibility of T depends only on the signs of such differences, which are constant on chambers. The example is checked in `check_conditionals.py`. □

Proposition 6.7 (Monotone cases)**.** *Assume* (f,g) *is nondegenerate and every cell defines a facet. Then* $X_{f,g}$ *is monotone if and only if* f *and* g *are uniform, and uniform marginals are nondegenerate exactly when* $\gcd(n,m) = 1$*. In particular, in the square case* $Y = Y'$ *of coupling 2-cells no nondegenerate pair is monotone. On the wall, the uniform* 2×2 *pair still gives a Delzant segment,* $X = \mathbb{CP}^1$*, with the product coupling as its midpoint; the uniform* $n \times n$ *pair with* $n \geq 3$ *gives a polytope that is not simple.*

Proof. For a Delzant polytope written with primitive normals, the toric manifold is monotone exactly when some interior point has equal lattice distance to all facets: the classes of the toric divisors span H^2 subject to the linear relations given by the normals, c_1 is the sum of the divisor classes, and $[\omega]$ is the sum weighted by the facet distances. Here the distances are the entries $\alpha(c)$, so they are equal exactly when α is constant, which forces f and g to be uniform; then $\alpha = f \otimes g$. Uniform marginals are nondegenerate iff $a/n \neq b/m$ for $0 < a < n$, $0 < b < m$, iff $\gcd(n,m) = 1$, which fails when $n = m$. The two wall cases are checked in `check_conditionals.py` (T3). □

Example 6.8 (The degree-6 del Pezzo surface)**.** For f uniform on two points and g uniform on three, $\mathrm{Cpl}(f,g)$ is a hexagon. In the basis of Λ above, its facet normals are $\pm e_1$, $\pm e_2$ and $\pm(e_1+e_2)$. This is the fan of $\mathbb{CP}^2$ blown up at three points, the del Pezzo surface of degree 6. All six entries of the product coupling equal $\frac{1}{6}$, so it is the equidistant point. The case 3×4 (uniform) gives a Delzant polytope of dimension 6 with 96 vertices and all 12 cells facets, again monotone.

Proposition 6.9 (Support floors are reduced margins)**.** *Let* $\tau \geq 0$ *on a set* D *of cells,* $f' = f - \tau \mathbf{1}_D \mathbf{1}$ *and* $g' = g - \mathbf{1}^\top \tau \mathbf{1}_D$*. Then* $\pi \mapsto \pi - \tau \mathbf{1}_D$ *is a bijection from* $\mathrm{Cpl}_\tau(f,g) = \{\pi \in \mathrm{Cpl}(f,g) : \pi \geq \tau \text{ on } D\}$ *onto* $\mathrm{Cpl}(f',g')$*. Hence* $\mathrm{Cpl}_\tau(f,g)$ *is non-empty iff* $f', g' \geq 0$*; it is a translate of a (rescaled) coupling polytope, so its normal fan, its toric type and its chamber are those of* (f',g')*, and its walls*

are $f'(I) = g'(J)$. *The largest common margin* δ *with* $\pi \geq \tau + w\delta$ *on* D *is* min *over rows and columns of (margin − floors)/(weights) whenever that value is non-negative.*

Proof. On D, $\pi \geq \tau$ is $\pi - \tau \mathbf{1}_D \geq 0$; off D the shift is zero; the margins shift by the row and column sums of $\tau \mathbf{1}_D$, and a transportation polytope with non-negative margins of equal total is non-empty. Replace τ by $\tau + w\delta$ for the last statement, valid while $\tau + w\delta \geq 0$. □

Example 6.10 (A policy tears the del Pezzo surface)**.** In example 6.8 put a floor τ on the single cell $(0, 0)$. The reduced margins are $(\frac{1}{2} - \tau, \frac{1}{2})$ and $(\frac{1}{3} - \tau, \frac{1}{3}, \frac{1}{3})$, and the two walls $f'_0 = g'_1$ and $f'_0 = g'_2$ are crossed together at $\tau = \frac{1}{6}$. The floored polytope is a hexagon for $\tau < \frac{1}{6}$ and a quadrilateral, of type $\mathbb{F}_1$, for $\frac{1}{6} < \tau < \frac{1}{3}$: the policy alone performs the two blow-downs of proposition 6.6. At $\tau = \frac{1}{3}$ the reduced margin g'_0 vanishes and the polytope loses dimension: the policy breaks the system. Floors on events $\pi(E) \geq \tau$, by contrast, give general polytopes. Exact vertices are in `check_support_floors.py`.

7 The imported layer: Floer theory of the special fibre

We now quote results from symplectic topology. They are not proved or checked here.

Nondisplaceability. For a monotone toric manifold, the torus fibre over the point at equal distance from all facets (the *special fibre*) is a stem in the sense of Entov and Polterovich, hence nondisplaceable [16]. For toric Fano manifolds, Cho and Oh show that a torus fibre with a flat $U(1)$ connection has nonvanishing Floer cohomology when it corresponds to a critical point of the disc potential [14]; see [13] for the Clifford torus and [17] for general compact toric manifolds.

Consequence for couplings. In the monotone cases of proposition 6.7, the special fibre lies over the product coupling $f \otimes g$. For example 6.8, the disc potential of the monotone special fibre is, up to the factor $T^{1/6}$, the Laurent polynomial

$$W(x, y) = x + y + \frac{1}{x} + \frac{1}{y} + xy + \frac{1}{xy},$$

read off from the facet normals, and it has a critical point at $(x, y) = (1, 1)$ (checked exactly), so the special fibre with trivial holonomy has nonzero Floer cohomology by [14].

The maximal coupling, by contrast, has zero entries whenever $f \neq g$, so it lies on the boundary of $\mathrm{Cpl}(f, g)$, where the torus fibre collapses (`check_conditionals.py`, T6). The first draft's conjecture that the critical points are the maximal couplings should therefore be replaced. The geometrically distinguished coupling is the product coupling, which is also the maximum-entropy coupling and the critical point of Guillemin's potential (proposition 6.5).

8 Stability conditions: a question

This subsection replaces a draft that proposed a Bridgeland stability condition whose central charge is the disc potential, whose heart is the committed states, and whose Harder–Narasimhan filtration is the obstruction tower. None of those identifications is well typed. A central charge is an additive map $K_0 \to \mathbb{C}$ and W is a function on a torus. Committed states are vertices of a polytope and form no abelian category. The obstruction tower assigns numbers to a claim; it does not filter an object. What can be said precisely is the following.

Bridgeland stability. A stability condition on a triangulated category $\mathcal{D}$ is a pair $(Z, \mathcal{A})$. Here $\mathcal{A}$ is the heart of a bounded t-structure, and $Z\colon K_0(\mathcal{D}) \to \mathbb{C}$ is additive, sends nonzero objects of $\mathcal{A}$ to the upper half-plane together with $\mathbb{R}_{<0}$, and admits Harder–Narasimhan filtrations; the support property is also required [11]. On any smooth projective surface X, and for any ample class ω, there is a stability condition with central charge

$$Z_\omega(E) = -\int_X e^{-i\omega}\,\mathrm{ch}(E)$$

whose heart is a tilt of $\mathrm{Coh}\,X$ [6, 21]. (A B-field can be added; we set it to zero.)

Proposition 8.1 (Marginals give stability conditions on the 2×3 surfaces)**.** *Let (f, g) be nondegenerate in the 2×3 family, let X_Σ be the toric surface of its chamber, and let D_c be the toric divisor of the facet $\{\alpha_c = 0\}$. Put*

$$[\omega_{f,g}] = \sum_{c \text{ facet}} f_i g_j\,[D_c], \qquad c = (i, j),$$

the product coupling read as divisor coefficients. Then:

(i) $[\omega_{f,g}]$ is the class of the toric symplectic form with moment polygon $\mathrm{Cpl}(f, g)$. It is independent of the base point used to write the polygon, and it is ample.

(ii) $\omega_{f,g} \cdot D_c$ is the lattice length of the edge $\{\alpha_c = 0\}$ of the coupling polygon, and $\frac{1}{2}\omega_{f,g}^2$ is its lattice area.

(iii) There is a stability condition $\sigma_{f,g} = (Z_{\omega_{f,g}}, \mathcal{A}_{\omega_{f,g}})$ on $D^b(\mathrm{Coh}\,X_\Sigma)$. It depends continuously on (f, g) within the chamber, and

$$Z(\mathcal{O}_X) = \mathrm{area}\,\mathrm{Cpl}(f, g), \qquad Z(\mathcal{O}_x) = -1, \qquad Z(\mathcal{O}_{D_c}) = \tfrac{1}{2}D_c^2 + i\ \mathrm{length}\{\alpha_c = 0\}.$$

(iv) As (f, g) approaches a wall from inside a chamber, one edge length tends to 0, its divisor E is a (-1)-curve, and $Z(\mathcal{O}_E) \to -\frac{1}{2}$, a point on the negative real axis.

Proof. (i) If a Delzant polygon is $\{u : \langle u, v_c\rangle + \lambda_c \geq 0\}$, the symplectic class is $\sum_c \lambda_c[D_c]$, up to the normalization of ω. Here the facet functions are $\alpha_c(u) = \langle u - u_0, v_c\rangle + \alpha_c(u_0)$ for any interior base point u_0, and choosing $u_0 = f \otimes g$ gives the formula. Another choice of base point changes the coefficients by $\langle w, v_c\rangle$, which is a linear relation among the $[D_c]$. A Kähler class on a projective toric surface is ample. (ii) These are standard toric identities: the degree of ω on D_c is the lattice length of the corresponding edge, and the symplectic volume is the lattice area (Duistermaat–Heckman). (iii) Apply [6, 21] with $\omega = \omega_{f,g}$, which varies continuously with (f, g). The values follow from $\mathrm{ch}(\mathcal{O}_X) = 1$, $\mathrm{ch}(\mathcal{O}_x) = [\mathrm{pt}]$ and $\mathrm{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$. (iv) By proposition 6.6 and `check_stability_layer.py` (S5), the vanishing edge is that of a (-1)-curve. All identities are checked in all 18 chambers. □

So the probabilistic data enter Bridgeland stability through the Kähler class, and nowhere else. The product coupling fixes the class, edge lengths of the coupling polygon are imaginary parts of central charges, and its area is $Z(\mathcal{O}_X)$. At the uniform point $\omega = -\frac{1}{6}K_X$ on the del Pezzo surface, with area $\frac{1}{12}$ and six edges of length $\frac{1}{6}$.

Questions.

(Q1) *Across walls.* Crossing a wall blows up or blows down a point (proposition 6.6), and Orlov's theorem gives $D^b(\mathrm{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X) \rangle$ [22]. As the wall is approached, $Z(\mathcal{O}_E)$ tends to the negative real axis. Does the path $\sigma_{f,g}$ extend across the wall to a path through stability conditions glued along Orlov's decomposition? If so, the chamber structure of the marginals becomes a wall structure in a single space of stability conditions.

(Q2) *Objects for couplings.* Is there a natural object $E_\alpha \in D^b(\mathrm{Coh}\, X)$ attached to an interior coupling α? A candidate is the SYZ or mirror transform of the torus fibre over α with a flat connection. Does $Z(E_\alpha)$ have a probabilistic meaning, for instance in terms of the Fisher metric of proposition 6.5?

(Q3) *Quantum corrections.* Iritani's integral structure identifies quantum-corrected central charges of objects of $D^b(\mathrm{Coh}\, X)$ with oscillatory integrals $\int e^{-W/z}$ of the mirror potential over integral cycles, and their asymptotics involve the $\widehat{\Gamma}$-class [20, 18]. This is the precise sense in which the disc potential is related to central charges: through oscillatory integrals, not through its values. Do the corrected central charges, with $\omega = \omega_{f,g}$, define stability conditions on these surfaces? Does the monotone point $f \otimes g$ uniform play a special role?

(Q4) *Higher dimensions.* For $(n-1)(m-1) \geq 3$ the toric manifolds have complex dimension at least three. There, the existence of stability conditions is known only in special cases, through Bogomolov–Gieseker type inequalities [10]. The construction of proposition 8.1 therefore extends only conditionally.

What is not claimed. We make no identification of the disc potential with a central charge, of committed states with a heart, of the obstruction tower with Harder–Narasimhan filtrations, or of hallucination with instability. The obstruction tower and the stability conditions above live on different objects (claims and admissibility data on one side, coherent sheaves on $X_{f,g}$ on the other), and any relation between them would have to be constructed. Hallucination keeps a precise meaning, but on the tower side: a claim whose required coherent lift is obstructed [4, Def. 10.17], a convex obstruction with a certificate. Unfillable horns are instances of it (theorem 11.2). Table 1 lists each withdrawn identification, the reason, and what is claimed instead.

9 The conjectural layer

Proposition 9.1 (The 2×3 family [9, §3])**.** *In the* 2×3 *family there are* 18 *chambers. Every toric surface* $X_{f,g}$ *is the degree-6 del Pezzo surface or a toric blow-down of it, and crossing one wall is exactly one toric blow-up or blow-down.*

Conjecture 9.2 (Family over the chamber complex)**.** *For general* $n \times m$*, the toric manifolds* $X_{f,g}$ *form a family over each chamber, and crossing a wall is a toric birational transformation (in higher dimension possibly a flip) determined by the pair* (I, J) *defining the wall. On a wall,* $X_{f,g}$ *degenerates to a singular toric variety whose moment polytope is the non-simple* $\mathrm{Cpl}(f, g)$*.*

Question 9.3 (Location of Floer-nontrivial fibres)**.** Outside the monotone cases, the fibres with nonvanishing Floer cohomology (possibly after bulk deformation) are located by the potential of [17]. Is the product coupling always among them, and do they have a statistical meaning?

Question 9.4 (Beyond the minimal cone)**.** Theorem 5.2 makes whiskering an A_∞-functor into the square-zero cone, with two components. (i) In the free cone, do the higher Boolean cumulants give the higher components? (ii) Does the construction extend to the multi-object setting, an A_∞-functor between linear categories whose objects are 1-cells, compatible with the nerve of section 4?

Conjecture 9.5 (Fukaya-type functor)**.** *There is a functor from a category built from the Fukaya categories of the $X_{f,g}$ (with special fibres as distinguished objects) to a linearization of the coupling 2-cells, compatible with gluing on one side and composition of Lagrangian correspondences on the other.*

10 Experimental definitions and evidence

This section takes up the five problems left open by the previous draft. Each subsection states a definition, proves what can be proved, and records what was only computed. The numbers are tabulated in [9, §13], and the scripts are listed in section 14.

10.1 The multi-object cone

Definition 10.1 (The linear category of couplings)**.** Let $\mathcal{L}$ be the category whose objects are full-support distributions f on finite sets. Its morphisms are

$$\mathcal{L}(f,g) = V(f,g) = \{a \in \mathbb{R}^{Y\times Y'} : a\mathbf{1} = s\,f,\ \mathbf{1}^{\mathsf{T}}a = s\,g^{\mathsf{T}} \text{ for some } s = s(a) \in \mathbb{R}\},$$

with composition $a \cdot b = a\,\mathrm{diag}(g)^{-1}b$ and identity $1_f = \mathrm{diag}(f)$. Let $\mathcal{I}(f,g) = \{a : s(a) = 0\}$ be the circulations, and let $\mathcal{C}$ be the mapping cone of $\mathcal{I} \hookrightarrow \mathcal{L}$: it has $\mathcal{C}(f,g) = V(f,g) \oplus \varepsilon\,\mathcal{I}(f,g)$, with $\deg\varepsilon = -1$, $m_1(\varepsilon x) = x$ and $\varepsilon x \cdot \varepsilon y = 0$. A family of kernels $h = (h_Y : Y \to Z_Y)$ acts on morphisms in two ways. The *product action* is $\pi_h(a) = h_Y^{\mathsf{T}} a h_{Y'}$, which runs h independently on both coordinates. The *whiskering* w_h is the postcomposition of theorem 2.3(ii): on $V(f,g)$ with f, g on the same set Y, it copies h_Y on equal pairs and runs it independently on unequal pairs; between different sets it is π_h. For $F_1^h \in \{\pi_h, w_h\}$ put $F_2^h = \varepsilon\,\delta_h$ with $\delta_h(a,b) = F_1^h(a \cdot b) - F_1^h(a) \cdot F_1^h(b)$, and $F_n^h = 0$ for $n \geq 3$.

Couplings are the morphisms with $a \geq 0$ and $s = 1$, and their composition is gluing. When $f = g$ this is the algebra A_f of section 5. For kernels $X \to Y$ as objects, take the product over $x \in X$.

Theorem 10.2 (Multi-object version of theorem 5.2)**.** *(i) $\mathcal{L}$ is a linear category, s is a functor to the one-object category $\mathbb{R}$, and its kernel $\mathcal{I}$ is a two-sided ideal with $\dim\mathcal{I}(f,g) = (|Y| - 1)(|Y'| - 1)$. Hence $H^0(\mathcal{C}) = \mathcal{L}/\mathcal{I}$ has every hom-space equal to $\mathbb{R}$, and $H^{-1}(\mathcal{C}) = 0$.*

(ii) For either choice of F_1^h, F^h is an A_∞-functor $\mathcal{L} \to \mathcal{C}$ with two components.

(iii) With the whiskering w_h it is strictly unital: $w_h(1_f) = 1_{h\#f}$ and $F_2^h(1_f, a) = F_2^h(a, 1_f) = 0$. With the product action π_h it is only cohomologically unital: $\pi_h(1_f) - 1_{h\#f} = m_1(\varepsilon u_f)$ with $u_f \in \mathcal{I}$ nonzero in general.

(iv) The product action is strictly functorial in the kernel: $F^{\pi,h'} \circ F^{\pi,h} = F^{\pi,hh'}$ as A_∞-functors, where $F^{h'}$ acts on $\varepsilon\mathcal{I}$ by $F_1^{h'}$. The whiskering is functorial for deterministic kernels, where $w_h = \pi_h$, but not for stochastic ones: $w_{h'}w_h \neq w_{hh'}$ in general.

Proof. (i) $(a\,\mathrm{diag}(g)^{-1}b)\mathbf{1} = s(b)\,a\mathbf{1} = s(a)s(b)f$, and similarly for column sums, so composition is well typed and s is multiplicative. The dimension count is that of the transportation polytope's tangent space. (ii) Both actions are linear, push the margins forward along h and preserve s, because kernels preserve mass, so δ_h and u_f have $s = 0$. The arity-two and arity-three equations are those in the proof of theorem 5.2, which use only linearity and associativity and never that source and target coincide. (iii) The whiskering sends the diagonal $\Delta_f = 1_f$ to $\Delta_{h^\# f}$, so $F_2^h(1_f, a) = w_h(a) - 1_{h^\# f} \cdot w_h(a) = 0$, and similarly on the right; the product action sends 1_f to $h^\mathsf{T}\,\mathrm{diag}(f)h$, which differs from $\mathrm{diag}(h^\# f)$ when h is not deterministic. (iv) π composes because $(hh')^\mathsf{T} a(hh') = h'^\mathsf{T}(h^\mathsf{T} ah)h'$, and the composite's second component $F_1^{h'}\delta_h(a,b) + \delta_{h'}(F_1^h a, F_1^h b)$ telescopes to $\delta_{hh'}(a,b)$. For w, an unequal pair (y, y') goes under w_h partly to equal pairs (z, z), which $w_{h'}$ then copies, while $w_{hh'}$ runs hh' independently on (y, y'). For a deterministic kernel $h(y) \otimes h(y)$ is the point mass at $(h(y), h(y))$, so $w_h = \pi_h$ and functoriality follows from that of π. `check_multiobject_cone.py` exhibits the failure for stochastic kernels. □

So no single action is both strictly unital and strictly functorial in stochastic kernels. Copying on equal pairs, which makes the diagonal couplings identities, is exactly what breaks composition of kernels.

Definition 10.3 (Nerve to cone)**.** For a 2-simplex p of the coupling nerve, with edges $d_2p \in V(f_0, f_1)$, $d_0p \in V(f_1, f_2)$ and $d_1p \in V(f_0, f_2)$, put $H(p) = \varepsilon\,(d_1p - d_2p \cdot d_0p) \in \varepsilon\,\mathcal{I}(f_0, f_2)$.

Proposition 10.4. *$m_1 H(p) = d_1p - d_2p \cdot d_0p$, so every 2-simplex is a homotopy from the composite of two of its edges to the third. H vanishes on gluing fillers and on degenerate 2-simplices. It is natural for transport along kernels, with the product action: $H(h_*p) = \pi_h(H(p)) + F_2^{\pi,h}(d_2p, d_0p)$, where h_*p pushes the 2-simplex forward along h in each coordinate. On a 3-simplex the coherence class $Z = \gamma_{01}H_{123} + H_{013} - H_{012}\gamma_{23} - H_{023}$ vanishes identically.*

Proof. The first two claims follow from the definition and theorem 4.2(iii). Naturality is the identity $h_*(d_1p) - h_*d_2p \cdot h_*d_0p = h_*(d_1p - d_2p \cdot d_0p) + \delta_h(d_2p, d_0p)$. Expanding d_1 of each face of a 3-simplex gives $m_1 Z = 0$, and m_1 is injective on $\varepsilon\mathcal{I}$. □

So question 9.4(ii) has a positive answer on the 2-skeleton. The answer is also limited: since $H^{-1}(\mathcal{C}) = 0$, the square-zero cone cannot see the 3-simplex obstructions of theorem 4.3. Detecting contextuality needs a target with nonzero H^{-1}, such as the free cone of question 9.4(i).

10.2 Wall-crossing as variation of GIT

Proposition 10.5 (Cox presentation and walls)**.** *$X_{f,g}$ is the GIT quotient $\mathbb{C}^{n\times m}/\!\!/_{(f,g)}(T^n \times T^m)/\Delta$, where the cell (i, j) has weight $e_i \oplus e_j$. The kernel of the ray map $\mathbb{Z}^{n\times m} \to \Lambda^*$, $c \mapsto v_c$, is the saturated lattice of matrices $r_i + s_j$, of rank $n + m - 1$. The hyperplanes in the character space $\{\sum f = \sum g\}$ spanned by weights are exactly the walls $\sum_I f = \sum_J g$ and the facets $f_i = 0$, $g_j = 0$ of the weight cone.*

Proof. The first statement is Cox's construction [33] for the fan of section 6, with the kernel checked in `check_vgit.py` (V1). For the second, a functional (a, b) vanishes on the weights of a set S of cells if and only if $a_i + b_j = 0$ for $(i, j) \in S$, that is, if and only if a and $-b$ are constant on the connected components of the bipartite graph S. The rank of S is $n + m$ minus the number of components. It is $n + m - 2$ exactly when there are two components, $I \sqcup J$ and $I^c \sqcup J^c$. Then the functional is $\sum_I f - \sum_J g$, or f_i or g_j when a component is a single isolated vertex. □

So conjecture 9.2 is an instance of variation of GIT for torus actions [35, 34]. The quotient is constant on chambers and changes by a birational transformation at each wall. What remains is the type of transformation. In the examples, every single-wall crossing is one toric blow-up or blow-down, or a flip with the same rays. For 2×4, 40 random crossings gave 18 blow-ups or blow-downs and 22 flips. For 3×3, 25 crossings gave 6 and 19 (`check_vgit.py` (V3)).

10.3 Across a wall

Follow the path $P(t)$ of [9, §12] from the pentagon chamber ($X_- = \mathrm{Bl}_2\mathbb{CP}^2$) to the hexagon chamber (X_+ the del Pezzo surface of degree 6). It crosses only the wall $f_1 = g_1$, at $t = \frac{3}{8}$. Let $\pi\colon X_+ \to X_-$ blow up the fixed point $D_{01} \cap D_{02}$, with exceptional curve $E = D_{10}$, and let $s(t) = f_1 - g_1$, the lattice length of the new edge.

Proposition 10.6 (Central charges across the wall)**.** *Let $\omega_-(t)$ be the pentagon class with its support numbers continued affinely past the wall. Then:*

(i) $\omega_+ = \pi^*\omega_- - sE$, *so* $\mathrm{area}_+ = \mathrm{area}_- - \frac{1}{2}s^2$. *In particular the jump $\frac{16}{225}$ of the second derivative of the area [9, §12] is $(ds/dt)^2$.*

(ii) $Z_+(\pi^*F) = Z_-(F) - \mathrm{rk}(F)\,\frac{1}{2}s^2$ *for all* $F \in K_0(X_-)$, *and* $Z_+(\mathcal{O}_E(-1)) = \frac{1}{2} + is$.

(iii) At the level of K_0, Orlov's decomposition $\langle \mathcal{O}_E(-1), \pi^*D^b(X_-)\rangle$ holds: $\chi(\mathcal{O}_E(-1), \mathcal{O}_E(-1)) = 1$, $\chi(\pi^*F, \mathcal{O}_E(-1)) = 0$, *and* $K_0(X_+) = \mathbb{Z}[\mathcal{O}_E(-1)] \oplus \pi^*K_0(X_-)$.

Proof. (i) π^* is an isometry onto $E^\perp$ with $E^2 = -1$, and the difference of support numbers on E is $h_{10} - h_{01} - h_{02} = g_1 - f_1$. (ii) ch commutes with π^*. Then $(\pi^*\omega - sE)\cdot\pi^*D = \omega\cdot D$ and $(\pi^*\omega - sE)^2 = \omega^2 - s^2$, and $\mathrm{ch}\,\mathcal{O}_E(-1) = (0, E, -\frac{1}{2})$. (iii) This is Riemann–Roch, checked on generators of $K_0(X_-)$. All of it is verified exactly along the path in `check_across_wall.py`. □

Definition 10.7 (Glued central charge)**.** Along the path, define Z_t on $K_0(X_+) = \mathbb{Z}[\mathcal{O}_E(-1)] \oplus \pi^*K_0(X_-)$ by $Z_t = Z_-$ on π^*K_0 for $t \le \frac{3}{8}$, and by Z_+ for $t > \frac{3}{8}$, with $Z_t(\mathcal{O}_E(-1)) = \frac{1}{2} + i\,s(t)^+$.

By proposition 10.6, Z_t is continuous at the wall, and the new class enters at the real point $\frac{1}{2}$: the phase of $\mathcal{O}_E(-1)$ tends to 0, and that of $\mathcal{O}_E$ tends to 1. This answers (Q1) of section 8 at the level of central charges. The glued charges extend continuously, in the form the gluing construction of [32] requires along Orlov's decomposition. The hearts and the gluing condition on them are not tested.

10.4 The Fukaya probe

Definition 10.8 (Candidate object map)**.** For u in the interior of $\mathrm{Cpl}(f, g)$, let $\mathfrak{PO}(u; w) = \sum_c \exp(-\lambda\alpha_c(u) + \langle n_c, w\rangle)$, with $T = e^{-\lambda}$. A critical point w whose real part stays bounded as $\lambda \to \infty$ makes the fibre $L(u)$ Floer-nontrivial, with holonomy $\mathrm{Im}\,w$ [17]. The candidate object map of conjecture 9.5 sends $(L(u), \mathrm{Im}\,w)$ to the coupling u.

Since $\mathfrak{PO}(u; w) = \mathfrak{PO}(0; w - \lambda u)$, the locations $u_k = -\lim \mathrm{Re}\,w_k/\lambda$ of the critical points of $\mathfrak{PO}(0; \cdot)$ are intrinsic.

Proposition 10.9 (Monotone point)**.** *At $f = (\frac{1}{2}, \frac{1}{2})$, $g = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the six critical points of W (example 6.8) all lie over the product coupling. The Hessian of W in logarithmic coordinates is a positive multiple ($\frac{1}{6}$) of the Fisher metric $\sum_c n_c n_c^{\mathsf{T}}/\alpha_c$ at the product coupling at the critical point $(1, 1)$, and at no other.*

Proof. All α_c equal $\frac{1}{6}$ at the product coupling, so $\operatorname{Hess} W(1,1) = \sum_c n_c n_c^{\mathsf{T}} = \frac{1}{6}\cdot\text{Fisher}$. At the other critical points the coefficients x^{n_c} are non-real or have mixed signs (`check_fukaya_probe.py`, K3). □

Observation 10.10 (Off the monotone point)**.** At $(f_1, g_1, g_2) = (0.52, 0.31, 0.35)$ in the hexagon chamber, the six critical points, tracked to $\lambda = 2560$, lie over three couplings. Four lie over the *leximin coupling*, which maximizes the smallest cell, then the next smallest, and so on. One lies over each of two couplings where three minimal cells balance with holonomy signs, $\sum_c \pm n_c = 0$. The product coupling is not among them. Off the monotone point the Hessian of $\mathfrak{PO}$ carries Boltzmann weights $e^{-\lambda\alpha_c}$, not the Fisher weights $1/\alpha_c$, and the two forms are not proportional. In `fukaya_scan.py`, which finds all critical points by elimination at $\lambda = 500$ and 2000, the same pattern holds at three further random points of the chamber. There are three locations with multiplicities $4 + 1 + 1$, and the leximin coupling carries four of them, including the positive real one.

Proposition 10.11 (The product fibre is generically Floer-trivial)**.** *In the hexagon chamber, suppose the smallest of the six numbers $f_i g_j$ is attained at a single cell. Then the fibre over the product coupling has no critical point of $\mathfrak{PO}$ with unitary holonomy, so its Floer cohomology vanishes (without bulk deformation).*

Proof. For the toric Fano surface $X_{f,g}$, $\mathfrak{PO}(u; y) = \sum_c T^{\alpha_c(u)} y^{n_c}$ exactly [14]. At the product coupling $\alpha_c = f_i g_j$, and if c_0 is the unique minimizing cell, the lowest-valuation term of $y\,\partial_y \mathfrak{PO}$ in a direction k with $n_{c_0}^k \neq 0$ is $T^{\alpha_{c_0}} n_{c_0}^k y^{n_{c_0}}$. For unitary y this term has nonzero leading coefficient, and every other term has strictly larger valuation, so the derivative cannot vanish. By [17], Floer cohomology of $L(u)$ with a unitary local system is nonzero exactly at such critical points. □

At the monotone point all six $f_i g_j$ are equal and proposition 10.9 applies instead. At the first sampled point the six values are distinct, so proposition 10.11 applies. This answers question 9.3 negatively for the product coupling, with a proof. The Floer-nontrivial fibres do have a statistical meaning: the leximin coupling, the most balanced coupling in the max–min sense, consistent with the iterated max–min construction of a nondisplaceable fibre in [17], together with signed-balanced couplings. The candidate object map of definition 10.8 therefore lands on these couplings, not on the maximum-entropy coupling. The metric compatibility hoped for in conjecture 9.5 holds only at the monotone point.

11 The obstruction tower and the nerve

The tower paper [4] attaches obstructions to a claim. Given admissibility data $S_1 \subseteq \mathsf{D}B$ and $S_2 \subseteq \mathsf{D}(S_1)$, it uses three kinds of set: local sets $L_1 = \operatorname{conv} S_1$ and $L_2 = \mathsf{D}(S_1)$, effective sets $E_2 = \operatorname{conv} S_2$ and $E_1 = \mu_B(E_2)$, and the fibre $\mathcal{F}(x_1) = \{\Pi \in L_2 : \mu_B \Pi = x_1\}$ of level-two lifts. The *primary obstruction* is the failure of the local test $x_1 \in L_1$. The *secondary obstruction* is the interval $I(x_1) = o_3(\mathcal{F}(x_1)) = [o_3^-, o_3^+]$, where $o_3(\Pi)$ is the ℓ^1 distance from Π to E_2 [4, Def. 10.11, Def. 10.14, Prop. 10.15]. This section states how these obstructions relate to layers III and IV and to the toric geometry. It also records which identifications fail.

Typing at level two. In a coherent context $(\rho, \Pi, \Omega, \Sigma)$ the first component is determined by the second: $\rho = \mu_B \Pi$. The pair (ρ, Π) is therefore a second-order state, not a coupling. Couplings enter between two contexts. For kernels f, g and a context c, the level-two grade of

proposition 3.4 is a Kantorovich distance, that is, a minimum over the level-two coupling polytope $\mathrm{Cpl}(A_f(c)_2, A_g(c)_2)$ of couplings of two second-order states. This polytope has dimension $(|\operatorname{supp} A_f(c)_2| - 1)(|\operatorname{supp} A_g(c)_2| - 1)$ and is Delzant when nondegenerate [9, Prop. 5.1]. Flattening maps it affinely to $\mathrm{Cpl}(\mu A_f(c)_2, \mu A_g(c)_2)$, and its fibres have fixed cross-covariance (proposition 5.1). So the toric geometry acts on pairs of transported contexts, not on a single context.

Definition 11.1 (A horn as a tower)**.** Let Λ be a horn on finite sets with faces $F_1, \dots, F_n$. Put $B = \bigsqcup_i Y^{F_i}$. Encode horn data $(p_i)_i$ as the level-one claim $x_1 = \frac{1}{n}\sum_i p_i$ in $\mathsf{D}B$, and a global assignment s as $\rho_s = \frac{1}{n}\sum_i \delta_{s|F_i}$. Let $\mathrm{Loc} \subseteq \mathsf{D}B$ be the encoded compatible families, the local polytope of [24]. Take $S_1 \subseteq \mathrm{Loc}$ finite and containing every ρ_s, and $S_2 = \{\delta_{\rho_s}\}$: only global mechanisms are effective.

Theorem 11.2 (Horn obstructions are secondary obstructions)**.** *In definition 11.1:*

(i) If S_1 is the vertex set of Loc*, then $L_1 = \mathrm{Loc}$, so the primary test is compatibility of the horn data.*

(ii) $E_1 = \mathrm{conv}\{\rho_s\}$, and $x_1 \in E_1$ if and only if the horn fills. An unfillable horn is an extension-obstructed claim in the sense of [4, Def. 10.17]: it passes the local test but has no coherent lift.

(iii) $o_3^-(x_1) \geq 2\,\mathrm{CF}(x_1)$, with equality when S_1 is the vertex set of Loc*. Here* CF *is the contextual fraction of theorem 4.3(iii). The inequality can be strict when S_1 is smaller: if S_1 consists of the ρ_s and the parity families, then $o_3^- = 2$ on the parity n-horn, while $2\,\mathrm{CF} = 2 - 2^{3-n}$.*

(iv) $d(x_1, E_1) \leq 2\,\mathrm{CF}(x_1)$ in ℓ^1.

(v) If $S_1 = \{\rho_s\}$, then $\mathcal{F}(x_1)$ is the polytope of fillers. For a 2-horn its directions are the slice-wise circulations of theorem 4.2(iii).

Proof. (i) Deterministic compatible families are exactly the ρ_s, and Loc is the convex hull of its vertices. (ii) $\mu_B(S_2) = \{\rho_s\}$. A global law G restricts to the data exactly when $\sum_s G(s)\rho_s = x_1$, because the faces sit in disjoint summands of B. (iii) $E_2 = \mathsf{D}(\{\rho_s\})$ inside $\mathsf{D}(S_1)$, so $o_3(\Pi) = 2\,\Pi(S_1 \setminus \{\rho_s\})$, as in [4, Prop. 10.9]. Minimizing over $\mathcal{F}(x_1)$ gives twice the least weight on non-global mechanisms over all decompositions of x_1 within S_1. This is at least the least w with $x_1 = (1-w)e + w\,e'$, $e \in \mathrm{conv}\{\rho_s\}$, $e' \in \mathrm{Loc}$. When S_1 contains all vertices of Loc, every such e' decomposes within S_1, so equality holds. The least w is CF: if $\sum_s \lambda_s \rho_s \leq x_1$, the remainder is non-negative and satisfies the linear compatibility equations, so the remainder constraint is automatic [23]. For the parity horns no decomposition within the smaller S_1 uses a global mechanism, which `check_tower_horns.py` confirms for $n = 3, 4, 5$. (iv) This is [4, Prop. 5.12], since $\mathrm{CF} = \mu_{\min}$ with the ρ_s as sources. (v) $\mu_B\Pi = x_1$ says that the global law Π restricts to the data, and then apply theorem 4.2. □

Corollary 11.3 (Tower levels are not horn dimensions)**.** *Every horn obstruction, of any dimension $n \geq 3$, appears at the same place in the tower: as an extension-obstructed level-one claim measured by the secondary obstruction. The primary test detects incompatible face data, not the failure of a 2-horn, since 2-horns always fill. Level three records uncertainty about the decomposition itself, a law on second-order states. The tower levels count orders of uncertainty, while the horn dimension counts the arity of a joint law. So the primary, secondary and tertiary obstructions do not correspond to 2-, 3- and 4-horns.*

For the parity horns, $d(x_1, E_1) = \frac{1}{3}, \frac{3}{8}, \frac{9}{20}$ for $n = 3, 4, 5$, which is twice the best approximate filler distance $\varepsilon^*(n)$ of [9, §13.1]. One inequality is general: since x_1 averages the face laws, $d(x_1, E_1) \le 2\varepsilon^*$. Equality is computed, not proved.

Proposition 11.4 (Information bounds on the obstructions)**.** *Let $y \in E_1$ have full support. Then*

$$d(x_1, E_1)^2 \le \|x_1 - y\|_1^2 \le \min\{\chi^2(x_1\|y),\, 2\,\mathrm{KL}(x_1\|y)\}, \qquad \chi^2(x_1\|y) = g^{\mathrm{F}}_y(x_1 - y, x_1 - y),$$

where g^{F} is the Fisher metric and KL *is the Bregman divergence of $\Phi(y) = \sum_b y(b) \log y(b)$, with* $\mathrm{Hess}\,\Phi = g^{\mathrm{F}}$*. The same bounds hold for $o_3(\Pi)$ with y replaced by any $\Pi' \in E_2$ of full support. The Fisher bound is an equality exactly when $|x_1 - y|$ is proportional to y. No reverse inequality holds, since* KL *is infinite when y misses part of the support of x_1.*

Proof. By Cauchy–Schwarz, $\big(\sum|v|\big)^2 = \big(\sum(|v|/\sqrt{y})\sqrt{y}\big)^2 \le \sum v^2/y$, with equality exactly when $|v|/\sqrt{y}$ is proportional to $\sqrt{y}$. Pinsker's inequality gives the KL bound. □

On a coupling polytope, $\Phi = \sum_c \alpha_c \log \alpha_c$ is twice the Guillemin potential of proposition 6.5. So obstruction distances between couplings are bounded by its Bregman divergence and by the Fisher form, and the circulations are the directions in which these are measured. In the tower example of [9, §11], with $x_1 = (\frac{7}{10}, \frac{3}{10})$ and $E_1 = \{(\frac{1}{2}, \frac{1}{2})\}$, the Fisher bound is attained: $\|x_1 - y\|_1^2 = \chi^2 = \frac{4}{25}$.

Proposition 11.5 (Transport can only lower the secondary obstruction)**.** *For a kernel $k\colon B \to B'$, transport the data to $k_\# S_1$ and S_2 pushed forward. Then $o_3^-(k_\# x_1) \le o_3^-(x_1)$ and $d(k_\# x_1, E_1') \le d(x_1, E_1)$.*

Proof. Pushforward along a kernel is an ℓ^1 contraction, pushes $\mathcal{F}(x_1)$ into $\mathcal{F}(k_\# x_1)$ and maps E_k onto E_k'. Take the minimizers before transport, push them forward, and compare. □

The tower paper runs these obstructions on synthetic deep ensembles, with a finite-sample test built from a certificate and the contraction of obstructions under label noise [4, Section 11]. A graded version of the level-two obstruction is a Kantorovich distance, a minimum over level-two coupling polytopes, and a claim obstructed at level two provably cannot learn [3, Lemma 52.1, Cor. 52.3].

A horn's nerve is a full simplex, so it is contractible, yet the parity horns are contextual [3, Prop. 3.10]: hypergraph acyclicity, not nerve homology, is the criterion of theorem 4.3.

What is not identified. The obstructions are numbers, values of linear programs, and their certificates are test functions on B, on S_1 or on assignments. Circulations are not obstructions. They are the directions of indeterminacy: tangent vectors of fibres and of filler polytopes, measured by the Fisher metric. Boolean cumulants are multilinear maps on composable couplings, while the tower's obstructions are attached to a single claim and indexed by order of uncertainty. No map sends one to the other, and the cumulant order is not the tower level. The monotonicity of failure levels [4, Thm. 10.5] and proposition 11.5 resemble properties of Harder–Narasimhan filtrations, but no filtration of an object is involved, and none is claimed.

12 A statistical reading

This section reads the earlier constructions with data. The propositions are proved. The numbers come from one seeded synthetic data set, tabulated in [9, §15] and generated by the script listed in

[4, Section 15]; the tower paper runs the obstructions of section 11 on synthetic deep ensembles in the same place.

The same constructions run on real data in [3]. Partial-identification bounds over all gluings cover held-out truth at rates 0.97–1.00 on media-bias corpora [3, §38.5], and there two-context extension polytopes are the filler polytopes of proposition 12.3 [3, §3.10].

Setting. A moderation model sees items with a label $Y \in \{\text{safe}, \text{harmful}\}$ and takes an action $A \in \{\text{accept}, \text{review}, \text{reject}\}$. Its confusion table, the joint law of (Y, A), is a coupling of the label marginal f and the action marginal g, so it is a point of $\mathrm{Cpl}(f, g)$. The synthetic truth has $f = (0.52, 0.48)$ and $g = (0.31, 0.35, 0.34)$, which lie in the hexagon chamber: the first point of observation 10.10. In this reading:

- the product coupling is a classifier at chance level, whose action is independent of the label;
- the vertices are the extreme confusion tables, supported on spanning trees;
- the range of one cell over $\mathrm{Cpl}(f, g)$ is its Fréchet interval $[\max(0, f_i + g_j - 1), \min(f_i, g_j)]$, the sharp partial-identification bound when only the marginals are known.

For example, $P(\text{harmful}, \text{accept})$ is only known to lie in $[0, 0.31]$, while its true value is 0.03.

Proposition 12.1 (Independence tests are toric)**.** *Let $\hat{\alpha}$ be the empirical table of N i.i.d. pairs, with full-support marginals $\hat{f}, \hat{g}$.*

(i) Pearson's statistic is $X^2 = N\, g^{\mathrm{F}}_{\hat{f}\otimes\hat{g}}(v, v)$ with $v = \hat{\alpha} - \hat{f} \otimes \hat{g}$. This is N times the Fisher metric of $\mathrm{Cpl}(\hat{f}, \hat{g})$ at its Birch point, evaluated on the empirical cross-covariance, which is a circulation (propositions 5.1 and 6.5).

(ii) The likelihood-ratio statistic is $G = 2N\ \mathrm{KL}(\hat{\alpha} \| \hat{f} \otimes \hat{g}) = 2N\, \hat{I}(Y; A)$, which is $2N$ times the Bregman divergence of $\Phi = \sum_c \alpha_c \log \alpha_c$, twice the Guillemin potential.

(iii) Under independence both are asymptotically χ^2 with $(n-1)(m-1) = \dim \mathrm{Cpl}(f, g)$ degrees of freedom.

Proof. (i) $X^2 = N \sum_c (\hat{\alpha}_c - \hat{f}_i \hat{g}_j)^2 / (\hat{f}_i \hat{g}_j)$, and the Fisher metric at α is $\sum_c v_c^2 / \alpha_c$. (ii) $D_\Phi(\alpha, \beta) = \sum \alpha \log(\alpha/\beta) - \sum(\alpha - \beta)$, and the second sum vanishes because both tables have total mass one. (iii) This is the classical result [5]. The number of free parameters of a table with fixed margins is the dimension of the transportation polytope. □

So the degrees of freedom of the test of independence are the complex dimension of the toric manifold $X_{f,g}$. In the synthetic data, $N = 2000$, $X^2 = 883.7$ and $G = 1026.7$. Under the product law the rejection rate at level 0.05 is 0.046 (Pearson) and 0.047 (G) over 4000 replications, which matches two degrees of freedom.

Proposition 12.2 (Identifying the chamber from samples)**.** *Let (f, g) be nondegenerate, with distance $\delta = \min_{(I,J)} |\sum_I f - \sum_J g|$ to the nearest of its $W = (2^{n-1} - 1)(2^m - 2)$ walls. From N i.i.d. pairs, the empirical marginals lie in the chamber of (f, g) with probability at least $1 - 2W \exp(-N\delta^2/2)$. Hence $N \geq 2\log(2W/\eta)/\delta^2$ identifies the chamber, and with it the toric surface $X_{f,g}$ and the stability conditions of proposition 8.1, with probability at least $1 - \eta$.*

Proof. For each wall, $\sum_I \hat{f} - \sum_J \hat{g}$ is the mean of the N independent variables $1[Y \in I] - 1[A \in J] \in [-1, 1]$, whose mean is $\sum_I f - \sum_J g$. By Hoeffding's inequality its sign is wrong with probability at most $2\exp(-N\delta^2/2)$. A union bound over the walls completes the proof. □

For the synthetic truth, $\delta = 0.13$ and $N \geq 649$ suffices at $\eta = 0.05$. For a pair at distance 0.02 from a wall it takes $N \geq 27{,}404$. The observed error rates are far below the bound [9, §15]. Near a wall, then, the geometry is statistically undetermined: the data cannot say which surface they are on.

Proposition 12.3 (Two models on one label set)**.** *Let two models A and B be scored on the same labels, with confusion tables α_A on $Y \times A$ and α_B on $Y \times B$.*

(i) The joint laws of (Y, A, B) that agree with both tables are the fillers of a 2-horn. By theorem 4.2 they form the product over y of transportation polytopes with margins $\alpha_A(y, \cdot)$ and $\alpha_B(y, \cdot)$.

(ii) The sharp bounds on any linear functional of the joint law, given the two tables, are its minimum and maximum over this polytope. Examples are ensemble accuracy and the probability that both models accept a harmful item.

(iii) The gluing filler is the joint law under conditional independence of A and B given Y, and it is the maximum-entropy filler. Every other filler differs from it by the conditional covariances $\mathrm{Cov}(1_{A=a}, 1_{B=b} \mid Y = y)$, which are slice-wise circulations.

Proof. This is theorem 4.2 with Y_1 the label. In (iii), the entries of a filler minus the gluing filler, on the slice $Y = y$ and divided by $P(Y = y)$, are the conditional covariances of the indicators. □

In the synthetic data, the two models make correlated errors on harmful items. The probability that both accept a harmful item is 0.019. Conditional independence predicts 0.003, and the two tables alone bound it by $[0, 0.030]$. So an evaluation that assumes conditionally independent errors, as a naive ensemble analysis does, understates this risk by a factor of six, and the confusion tables cannot rule the larger value out.

Remark 12.4 (Sampling noise versus contextuality). Horn data collected in separate studies, one table per face, are never exactly compatible. The incompatibility is a primary obstruction in the sense of theorem 11.2(i), and it shrinks like $N^{-1/2}$. The contextual fraction is a population quantity, a secondary obstruction. Its plug-in estimate converges, but it is biased upward in small samples. In the synthetic horn, with population contextual fraction 0.207, the mean estimate is 0.294 at $N = 100$ per study and 0.209 at $N = 10^5$. So compatibility should be tested, or the data projected onto Loc, before a contextual fraction is reported. On real data the same floor appears: four MBIC ideology groups give a median contextual fraction of 0.11 from about 240 sentences per context and 0 from the joint table [3, §38.8].

13 Next steps

The companion [9] computes every layer explicitly. Of the five problems left open by the previous draft, section 10 settles or reduces each:

(1) *Across a wall:* the central charges glue continuously (proposition 10.6). Open: the hearts, and the gluing condition of [32] on them.

(2) *General wall-crossing:* conjecture 9.2 is toric VGIT (proposition 10.5). Open: prove that a single wall is always a blow-up, a blow-down or a flip, as in all computed 2×4 and 3×3 crossings, and read off which from (I, J).

(3) *Higher horns:* the parity horns have contextual fraction $1 - 2^{2-n}$ for every n (theorem 4.3). Open: a closed form for the best approximate fillers $\varepsilon^*(n)$, which increase and stay below $\frac{1}{2}$ for $n \leq 8$.

(4) *Multi-object cone:* positive on the 2-skeleton (theorem 10.2 and proposition 10.4); the square-zero cone is homotopically trivial, so homotopy-invariant information is only the mass [1]. Open: a target with $H^{-1} \neq 0$ that sees 3-simplex obstructions, which is question 9.4(i).

(5) *The tower:* horn obstructions are secondary obstructions of encoded claims (theorem 11.2). Open: a proof that $d(x_1, E_1) = 2\varepsilon^*$ for symmetric horns, and an encoding of level three.

(6) *Fukaya:* the product fibre is generically Floer-trivial (proposition 10.11); at the sampled points the Floer-nontrivial fibres lie over leximin and signed-balanced couplings, and the Fisher metric matches only at the monotone point (proposition 10.9, observation 10.10). Open: a proof that leximin always carries the trivial-holonomy critical point, and the functor on morphisms.

14 Computational checks

All checks pass.

- `check_glue.py` (30 checks): proposition 2.1 and theorem 2.3.
- `check_stochastic_basechange.py` (13 checks): theorem 3.1; the hexagon of example 6.8; the Delzant property for random nondegenerate 3×3 marginals; non-simplicity of the uniform 3×3 case; the Hessian identity and critical point of proposition 6.5.
- `check_conditionals.py` (7 checks): theorem 6.4, propositions 6.6 and 6.7, and example 6.8.
- `check_stability_layer.py` (6 checks): proposition 8.1 in all 18 chambers of the 2×3 family.
- In the companion [9]:
  - `check_tower_fibration.py` (7 checks): theorem 3.1 and proposition 3.4;
  - `check_nerve.py` (10 checks): theorems 4.2, 4.3 and 5.2 and proposition 4.5;
  - `check_cumulants.py` (8 checks): proposition 5.1;
  - `check_polytope_companion.py`, `check_family.py` and `check_family_geometry.py`: proposition 9.1.
- For section 10:
  - `check_horns_n.py` (4 checks): theorem 4.3(iii) for all n, and $\varepsilon^*(n)$ for $n \leq 8$;
  - `check_vgit.py` (4 checks): proposition 10.5 and the crossing types;
  - `check_multiobject_cone.py` (11 checks): theorem 10.2 and proposition 10.4;
  - `check_across_wall.py` (7 checks): proposition 10.6;
  - `check_fukaya_probe.py` (6 checks) and `fukaya_scan.py`: proposition 10.9, observation 10.10.
  - `check_tower_horns.py` (9 checks): theorem 11.2, corollary 11.3, and propositions 11.4 and 11.5.
  - `synthetic_examples.py` (8 checks, with CSV data): the numbers of section 12. The CSV files are archived at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/blob/main/coupling/synthetic_data.zip`.

`check_support_floors.py` checks proposition 3.6 through the tower's floor results, theorem 4.8 (closed form against a linear program on 300 random families, and the antipodal sums against the global tables, exactly), example 4.9 and example 6.10 in exact arithmetic. All checks pass.

A Corrections to the first draft

Table 1: Withdrawn identifications, the reason, and what is claimed instead.

Withdrawn	Why	Claimed instead
Disc potential W as a central charge	A central charge is an additive map $K_0 \to \mathbb{C}$; W is a function on a torus	The central charge Z_ω of the product-coupling class (proposition 8.1). W enters only through the oscillatory integrals $\int e^{-W/z}$ of (Q3)
Committed states as a heart	A heart is an abelian category; committed states are vertices of a polytope	The tilted heart of $\mathrm{Coh}\, X_{f,g}$ (proposition 8.1); committed states are torus-fixed points
Obstruction tower as a Harder–Narasimhan filtration	An HN filtration is a chain of subobjects of one object, with decreasing phases; the tower assigns numbers to a claim	A convex obstruction theory with certificates; its monotonicity [4, Thm. 10.5] and proposition 11.5 are not a filtration
Hallucination as instability	It depended on the three identifications above	Hallucination as obstructed lifting [4, Def. 10.17]; unfillable horns are instances (theorem 11.2)
$F_2 = \mathrm{Hess}\, W$; Picard–Lefschetz formula	F_2 is bilinear on composable couplings with values in circulations; $\mathrm{Hess}\, W$ is a quadratic form at a critical point of a function on a torus. The cited check compared 0.1152 with 2, and no vanishing cycle was linked to composition	$F_2 = \varepsilon b_2$, a Boolean cumulant (theorem 5.2). $\mathrm{Hess}\, W$ matches the Fisher metric only at the monotone point (proposition 10.9)
The m_3 conjecture	The whiskering defect is the second component of a functor, not a ternary composition (item 5 below)	Whiskering is a two-component A_∞-functor (theorem 5.2); higher components are question 9.4(i)
22 chambers in the 2×3 family	A coordinate-order bug in the vertex code	18 chambers in 5 orbits [9, Thm. 3.1]
Cycle counts of $K_{n,n}$	The formula undercounted by a factor k; facets are cells, not cycles	$N_{2k} = \binom{n}{k}^2 k!(k-1)!/2$; $K_{4,4}$ has 204 cycles [9, App. A]
Bridgeland's construction for all surfaces	[12] treats K3 surfaces only	Arcara–Bertram [6] for any smooth projective surface and ample class

The items below record the remaining corrections to the first draft.

1. *Candidate disc potential.* $W_\beta(f) = \sum_g \int_{\mathrm{Cpl}(f,g)} e^{-\beta c(\alpha)}\, d\nu$ sums over a continuum of 1-cells g and is not defined as written. Its Laplace limit $\min_{g,\alpha} c(\alpha)$ is always 0, attained at $g = f$, $\alpha = \Delta_f$, so it carries no information. The toric disc potential of $X_{f,g}$ is the correct object (section 7).

2. *Symplectic structure.* A symplectic form is antisymmetric and the Fisher metric is symmetric, so the restriction of the one cannot be the other. The correct statement is proposition 6.5: the

Fisher metric is twice the canonical Kähler metric on the real section of $X_{f,g}$.

3. *Critical points.* "Critical points of $\pi\colon E \to \mathrm{Kl}(\mathsf{D})$" is undefined, since π is a functor, not a smooth map. The replacement is the special fibre over the product coupling in the monotone cases, and the walls of definition 6.3 as the locus where the family degenerates.

4. *Monodromy and Dehn twists.* The family $(f, g) \mapsto X_{f,g}$ is constant on chambers and changes by toric birational modifications across walls (proposition 6.6); no vanishing cycle or Dehn twist has been identified. Conjecture 9.2 restates the question.

5. *The m_3 conjecture.* Gluing is strictly associative (theorem 2.3(i)), so each hom-category, regarded as an A_∞-category, has $m_k = 0$ for $k \geq 3$. The whiskering defect compares $u \cdot (\beta \cdot \alpha)$ with $(u \cdot \beta) \cdot (u \cdot \alpha)$. Its inputs are two 2-cells and a 1-cell, and it measures the failure of whiskering to be a functor. In A_∞ language that is the second component $\mathcal{F}_2$ of an A_∞-functor, not a ternary composition (now theorem 5.2).

6. *Homotopy category.* The graded shadow has hom-sets $[d(f, g), \infty)$ under addition. These are not the degree-zero cohomology of cochain complexes, so the shadow is not a homotopy category of an A_∞-category without further structure.

7. *Fisher versus Fisher–Rao.* These are the same metric (proposition 6.1).

8. *References.* The companion tower is now version 9; its posted earlier version has the title "... and Weak Distributive Laws", a claim withdrawn in the revisions.

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